Document II: Background Questions

Participant identifier: player_253

1. **Specify your gender (mark one)**

   ☐ Female ☑ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. **Specify your level of education (mark one)**

   ☐ Undergraduate Student

   ☐ Master Student

   ☑ PhD Student

   ☐ PhD

   ☐ Other _____

3. **You currently work in...**

   ☑ Academia

   ☐ Industry

   ☐ Research center

   ☐ Student

   ☐ Other _____

4. **State your experience (in years) with software development** 14

5. **State your experience (in years) with modeling languages** 6

6. **I was familiar with the Blocky Maze game prior to this experiment:** ☐ Yes ☑ No

Document III: Perception Questionnaire

Participant identifier: Player_253

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	**5**
2. I had enough time to complete the tasks	1	2	3	4	**5**
3. I revised my models many times	1	2	**3**	4	5
4. I am satisfied with my final models	1	2	3	4	**5**
5. I lacked information or guidance while modeling	**1**	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	**4**	5
2. I found the environment unnecessarily complex	**1**	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	**5**
4. I think that I would need the support of a technical person to be able to use this environment	1	2	**3**	4	5
5. I found the various functions in this environment were well integrated	1	2	3	**4**	5
6. I thought there was too much inconsistency in this environment	1	**2**	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	**4**	5
8. I found the environment very cumbersome to use	1	**2**	3	4	5
9. I felt very confident using the environment	1	2	3	4	**5**
10. I needed to learn a lot of things before I could get going with this environment	**1**	2	3	4	5

3. Suggestions of the tool

1. I thought specifying the intent to the tool was tedious ..... [1] [2] [3] [4] [5]
2. I found that this tool helped me finding models matching my intent.. [1] [2] [3] [4] [5]
3. I felt that the candidates generated from the tool were not useful .... [1] [2] [3] [4] [5]

4. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

I started by droging the elements from the toolbar and then see what this choice will lead the robot and then I try to continue like this till reaching the goal.

2. **Which part of the task was most difficult, and why?**

of course the optimization task and being sure that there is no better way to solve the puzzle.

3. **Describe a moment when using the tool was helpful. Please detail as much as possible.**

In one of the stepts (step 8 or 9), I could not come up with a quick solution so I used the tool to find a first answer and then I optimized the proposed solution by editing the suggested model.

4. **Describe a moment when using the tool was irrelevant or misleading. Please detail as much as possible.**

In the lower steps (levels) the tool can be irrelevant because the maze can be solved quickly with a few steps.

Document II: Background Questions

Participant identifier: 601

1. **Specify your gender (mark one)**

   ☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. **Specify your level of education (mark one)**

   ☐ Undergraduate Student

   ☐ Master Student

   ☒ PhD Student

   ☐ PhD

   ☐ Other _____

3. **You currently work in...**

   ☒ Academia

   ☐ Industry

   ☐ Research center

   ☐ Student

   ☐ Other _____

4. **State your experience (in years) with software development** 17

5. **State your experience (in years) with modeling languages** 15

6. **I was familiar with the Blocky Maze game prior to this experiment:** ☐ Yes ☒ No

Document III: Perception Questionnaire

Participant identifier: 601

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	5
2. I had enough time to complete the tasks	1	2	3	4	5
3. I revised my models many times	1	2	3	4	5
4. I am satisfied with my final models	1	2	3	4	5
5. I lacked information or guidance while modeling	1	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	4	5
2. I found the environment unnecessarily complex	1	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this environment	1	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this environment	1	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	5
8. I found the environment very cumbersome to use	1	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this environment	1	2	3	4	5

3. Suggestions of the tool

1. I thought specifying the intent to the tool was tedious ..... [1] [2] [3] [4] [5]
2. I found that this tool helped me finding models matching my intent.. [1] [2] [3] [4] [5]
3. I felt that the candidates generated from the tool were not useful .... [1] [2] [3] [4] [5]

4. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

I tried to find patterns ~~xxxxxx~~ in parts of the problem to reuse these patterns in e.g. a loop

Starting from these patterns, I tried optimize further by finding more patterns or even reducing patterns to sequential steps

2. **Which part of the task was most difficult, and why?**

Knowing the distance of a "move forward" step was not always clear

Since the goals are conflicting there are many optimized solutions possible

3. **Describe a moment when using the tool was helpful. Please detail as much as possible.**

When the program did not work as intended, the forecasting of the path is helpful to narrow down the problem

4. **Describe a moment when using the tool was irrelevant or misleading. Please detail as much as possible.**

Running the program "cost" time watching the visualization even when the result is obvious due to the forecasting of the path

Document II: Background Questions

Participant identifier: Player_476

1. Specify your gender (mark one)

☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. Specify your level of education (mark one)

☐ Undergraduate Student

☐ Master Student

☒ PhD Student

☐ PhD

☐ Other _____

3. You currently work in...

☒ Academia

☐ Industry

☐ Research center

☐ Student

☐ Other _____

4. State your experience (in years) with software development 20

5. State your experience (in years) with modeling languages 5

6. I was familiar with the Blocky Maze game prior to this experiment: ☐ Yes ☒ No

Document III: Perception Questionnaire

Participant identifier: player_476

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	**5**
2. I had enough time to complete the tasks	1	2	3	4	**5**
3. I revised my models many times	1	2	**3**	4	5
4. I am satisfied with my final models	1	**2**	3	4	5
5. I lacked information or guidance while modeling	**1**	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	**2**	3	4	5
2. I found the environment unnecessarily complex	1	**2**	3	4	5
3. I thought the environment was easy to use	1	2	3	4	**5**
4. I think that I would need the support of a technical person to be able to use this environment	**1**	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	**4**	5
6. I thought there was too much inconsistency in this environment	**1**	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	**5**
8. I found the environment very cumbersome to use	**1**	2	3	4	5
9. I felt very confident using the environment	1	2	3	**4**	5
10. I needed to learn a lot of things before I could get going with this environment	**1**	2	3	4	5

3. Suggestions of the tool

1. I thought specifying the intent to the tool was tedious ..... [1] 2 3 4 5
2. I found that this tool helped me finding models matching my intent.. 1 2 [3] 4 5
3. I felt that the candidates generated from the tool were not useful 1 2 3 4 5

I did not try them.

4. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

~~I looked at the~~ sort of algorithm design, I see how can I reach my goal with min number of steps given the tools I have.

2. **Which part of the task was most difficult, and why?**

the tasks were not hard, but I felt missing some tools. In my head I would imagine a way to solve the task, I need a certain effects, but I could not apply since the tool is not available. for example for the last task it would be nicer to switch direction every time I need move left, right, ... but given the tools I got it was hard.

3. **Describe a moment when using the tool was helpful. Please detail as much as possible.**

Given my background I think it won't be helpful for me, but I imagine for some learning modeling ~~it~~ the visual effect is away nice to understand and imagine the solution.

4. **Describe a moment when using the tool was irrelevant or misleading. Please detail as much as possible.**

I think irrelevant or mis leading is not the terms I would, I think it just need more function ability to use more constructs like count, switch ...

Document II: Background Questions

Participant identifier: 181

1. Specify your gender (mark one)

☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. Specify your level of education (mark one)

☐ Undergraduate Student

☐ Master Student

☒ PhD Student

☐ PhD

☐ Other _____

3. You currently work in...

☒ Academia

☐ Industry

☐ Research center

☐ Student

☐ Other _____

4. State your experience (in years) with software development 10

5. State your experience (in years) with modeling languages 5

6. I was familiar with the Blocky Maze game prior to this experiment: ☒ Yes ☐ No

Document III: Perception Questionnaire

Participant identifier: 181

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	5
2. I had enough time to complete the tasks	1	2	3	4	5
3. I revised my models many times	1	2	3	4	5
4. I am satisfied with my final models	1	2	3	4	5
5. I lacked information or guidance while modeling	1	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	4	5
2. I found the environment unnecessarily complex	1	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this environment	1	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this environment	1	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	5
8. I found the environment very cumbersome to use	1	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this environment	1	2	3	4	5

3. Suggestions of the tool

1. I thought specifying the intent to the tool was tedious ..... [1] [2] [3] [4] [5]
2. I found that this tool helped me finding models matching my intent.. [1] [2] [3] [4] [5]
3. I felt that the candidates generated from the tool were not useful .... [1] [2] [3] [4] [5]

4. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

1. Identify the turns needed
2. Dependent on needed turns, choose either looped action pattern ~~for~~ or simple moves and turns to get to goal state

2. **Which part of the task was most difficult, and why?**

Keeping track of the power and memory objective when thinking about alternative solutions

3. **Describe a moment when using the tool was helpful. Please detail as much as possible.**

The simulation (live) helped a lot to understand the created solutions, including the animations

4. **Describe a moment when using the tool was irrelevant or misleading. Please detail as much as possible.**

- Too many buttons with unclear intent, e.g., "Skip end"
- All solutions found by search tool did never reach the goal state.
- The orange preview path of the solution seemed wrong/stale sometimes

Document II: Background Questions

Participant identifier: 735

1. Specify your gender (mark one)

☐ Female ☒ Male ☐ Not listed: _____ ☐ Prefer Not to Answer

2. Specify your level of education (mark one)

☐ Undergraduate Student

☐ Master Student

☒ PhD Student

☐ PhD

☐ Other _____

3. You currently work in...

☒ Academia

☐ Industry

☐ Research center

☐ Student

☐ Other _____

4. State your experience (in years) with software development 10

5. State your experience (in years) with modeling languages 0

6. I was familiar with the Blocky Maze game prior to this experiment: ☐ Yes ☒ No

Document III: Perception Questionnaire

Participant identifier: 735

1. Questions about the experiment

	strongly agree → ← strongly disagree				
1. The modeling tasks were easy to complete	1	2	3	4	5
2. I had enough time to complete the tasks	1	2	3	4	5
3. I revised my models many times	1	2	3	4	5
4. I am satisfied with my final models	1	2	3	4	5
5. I lacked information or guidance while modeling	1	2	3	4	5

2. Usability of the environment

In the following questions, the word *"environment"* refers to the **modeling environment you used to complete the Block game.**

1. I think that I would like to use this environment frequently	1	2	3	4	5
2. I found the environment unnecessarily complex	1	2	3	4	5
3. I thought the environment was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this environment	1	2	3	4	5
5. I found the various functions in this environment were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this environment	1	2	3	4	5
7. I would imagine that most people would learn to use this environment very quickly	1	2	3	4	5
8. I found the environment very cumbersome to use	1	2	3	4	5
9. I felt very confident using the environment	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this environment	1	2	3	4	5

3. Suggestions of the tool

1. I thought specifying the intent to the tool was tedious ..... [1] [2] 3 4 5
2. I found that this tool helped me finding models matching my intent.. [1] 2 3 4 5
3. I felt that the candidates generated from the tool were not useful .... 1 [2] 3 4 5

4. Open questions

1. **How did you approach the modeling task? Describe the steps you followed.**

① Visually look for repetitions in the maze (e.g., only right / left turns yield result).
② Add "repeat until" loop.
③ Specify condition blocks in loop.
④ Add turn & move statements.
⑤ Check if multiple move statements could be used to reduce power consumption.

2. **Which part of the task was most difficult, and why?**

The last maze was conceptually a lot more challenging than the others. Beyond that, rather straightforward.

3. **Describe a moment when using the tool was helpful. Please detail as much as possible.**

I used it as a sanity check once I found a solution. If the tool ~~would~~ could not generate a better solution in terms of power and memory. If not, it raised my confidence that my solution was close to optimal.

4. **Describe a moment when using the tool was irrelevant or misleading. Please detail as much as possible.**

I did not use it alot, except for checking. In hindsight, it wasn't fully clear to me, whether adding a solution from the tool would remove what I had already come up with.